# FRC 9470 2026 REBUILT

## SUMMARY

REBUILT challenged teams to compete in an industrial-themed game where alliances collected and launched “fuel” (5.91-inch foam balls) into their alliance’s hub to score points. Robots navigated field obstacles, adapted to alternating active and inactive hubs, and worked with human players to collect and deliver fuel throughout the match. The goal for both of our robots this year was to maximize fuel-scoring efficiency with a full wide intake and shooter.

## Robot 1 for Half Moon Bay and Silicon Valley District 2026

- Used WCP Swerve X2Ts to optimize our robot’s design; these modules’ profiles allowed us to have more space inside the robot for electronics and mechanisms. It featured a 25 inch wide intake that could pick up four balls at once. The hopper floor was fully covered in 1.25 inch rollers that allowed us to move fuel quickly through the robot and into the shooter. The shooter was a 3-laned shooter which allowed for maximum throughput of fuel. This allowed for us to shoot faster than most other teams with a turret with a relatively more simple design, involving very little moving parts.
- Ranked 3rd overall and was first pick of Alliance 1 at Half Moon Bay, advancing to finals and winning the event as well as winning the Excellence in Engineering award
- Ranked 2nd overall and was first pick of Alliance 1 at Silicon Valley, advancing to finals and winning the event as well as winning the Industrial Design award
- For this robot, I led the design process, creating the initial full concept sketch and mainly focusing on the shooter design. I also helped schedule the entire build process and manage the order list spreadsheet.
- This event was a huge milestone for our team as it was our first two competition wins and helped build momentum for the rest of the season.

# Robot 2 for Norcal Districts Championship, World Championships and Einstein

- Rebuilt the entire robot from scratch using the SDS mk5n swerve base. Rebuilt intake to be more robust and easily repairable. Revised hopper walls to be more robust and rigid. Changed hopper floor to use compliant wheels in the middle with lighter carbon fiber rollers everywhere else to reduce weight and move balls faster. Allowed the hopper floor to rotate so electrical systems are easier to access. Fully rebuilt shooter into the full wide “drum” to maximize ball output.
- Ranked 2nd and captained Alliance 2 at Norcal Districts Championship and became finalists at the event.
- DCMP debrief
    - Throughout the competition, the robot was falling apart especially the intake as it was made of polycarbonate and had already gone through multiple hours of practice
    - Hopper ceiling that keeps balls in the robot was also causing problems when trying to go under the trench
    - Shooter hood used for aiming kept vibrating so shot accuracy was worse
- Worlds revisions
    - Fully rebuilt intake to use double stacked .25” thick SRPP to maximize toughness and made intake basically indestructible
    - Added a slot with a bolt and bearing to the shooter hood to improve rigidity
- Ranked 2nd at World Championships Archimedes Division and joined Alliance 1, winning the division.
- Went 1-2 on Einstein Field
- Overall the best way to end off my time in FIRST.